The Value of Delayed Information: Predictive-Information Envelopes for Age-Limited Decisions

Envelopes, exact laws, and witnesses for age-limited decisions

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Abstract

A controller that acts on information of age D — a sensing delay, a telemetry lag, a stale cache — can outperform the best static policy only to the extent that the delayed observation still *predicts* the current state. Our companion systems work [1] proves this exactly for symmetric two-state Markov capacity ($\Delta(D) = \frac{1}{2}r(D)$, the autocorrelation at the sensing lag) with a β-mixing converse. This paper supplies the general theory. We (i) isolate the exact object — the *Bayes envelope* $\Delta_Y = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu})$, the expected excess value of posterior over prior beliefs; (ii) prove the universal *predictive-information envelope* $0 \le \Delta_Y \le \Omega\sqrt{\frac{1}{2}I(X;Y)}$ for any bounded objective with oscillation Ω, by Pinsker's inequality applied to the posterior process, together with its *incremental* refinement — the value of a richer delayed observation over a coarser one is bounded by $\Omega\sqrt{\frac{1}{2}I(X;H \mid Z)}$, so incremental decision value certifies *conditional* predictive information; (iii) derive the *bit-budget corollary*: a telemetry path delivering b bits per decision caps feedback value at $\Omega\sqrt{(b\ln 2)/2}$ regardless of what is encoded — compression can preserve but never create feedback value; (iv) show the envelope is *tight to leading order* in the canonical case: for the symmetric binary chain $I = \frac{r^2}{2} + O(r^4)$, so the bound reads $\Omega(\frac{|r|}{2} + O(|r|^3))$ against the exact $\frac{1}{2}|r|$; (v) prove by a two-bit construction that no universal equality $\Delta = f(I)$ can hold: mutual information is a scalar, decision value is a geometry, and two observations with identical I can carry all or none of the decision-relevant distinction; and (vi) give an exactness theory for binary state in which the local decision geometry at the prior sets the correlation exponent: for finite action sets the value of a delayed sample is exactly a sum of *hinges* in the lag covariance with sign-dependent dead zones; for twice-smooth objectives the weak-correlation response is quadratic to leading order, and for squared error exactly $C^2/[\pi(1-\pi)]$; the companion's linear law is the prior-on-kink polyhedral case. A nine-series battery (acoustics, bioacoustics, space weather, seismology, plus synthetic controls) finds no detectable history excess (equivalence-bounded within a registered margin) on the series where lag-1-matched surrogates predict none, observes the envelope's predicted $O(r^2)$ tightness on real data, and inverts the incremental theorem into model-calibrated lower bounds on conditional history information where the processes are non-Markov.

1 Introduction

Everywhere a controller acts on stale state — a scheduler admitting jobs against capacity sensed seconds ago, a trader on delayed quotes, a relay on an outdated channel estimate — the same question recurs: *what is the value of information of age D?* The age-of-information literature [7] measures freshness; freshness, however, has no decision semantics: a maximally fresh sample of an i.i.d. process is worthless, and a stale sample of a slowly-mixing one is nearly as good as the present. Our companion systems paper [1] made this precise in one setting, proving that for symmetric two-state Markov capacity the advantage of age-D feedback over the best static policy is *exactly* $\Delta(D) = \frac{1}{2}r(D)$ — the autocorrelation at the sensing lag — and measuring the law on a live GPU cluster. This paper supplies the theory that result belongs to. The thesis is one line:

Age matters only through predictive information, and predictive information matters only through the Bayes envelope.

Concretely: (**C1**) we isolate the exact object — the Bayes envelope $\Delta_Y = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu})$, classical in decision theory [2, 4, 3] — and prove the universal *predictive-information envelope* $\Delta_Y \leq \Omega\sqrt{I(X;Y)/2}$ (Theorem 2), with the *bit-budget corollary*: b bits of telemetry per decision cap feedback value at $\Omega\sqrt{(b \ln 2)/2}$, so compression can preserve but never create feedback value. (**C2**) We show the envelope is *sharp where it matters*: on the canonical binary chain it matches the exact law to $O(r^3)$ (Proposition 6) — and the battery of §7 *observes* this tightness on real data, including the predicted $O(r^2)$ correction. (**C3**) We prove no universal equality $\Delta = f(I)$ exists for bounded objectives (Proposition 7): information is a scalar, value is a geometry. This sharpens against the classical exception — for *logarithmic* (unbounded) utility, Kelly and Barron–Cover [5, 6] show the value of side information *equals* I. For general bounded objectives, mutual information provides a universal envelope but does not determine value; logarithmic utility is a distinguished objective for which equality is possible. (**C4**) We give an exactness theory for binary state whose organizing fact is that *the local geometry of the value function at the prior sets the correlation exponent.* For finite action sets (polyhedral value functions) the value of a delayed sample is exactly a sum of *hinges* in the lag-D covariance, with sign-dependent dead-zone thresholds — linear response beyond, identically zero within; for twice-smooth objectives the weak-correlation response is quadratic to leading order (exactly $C(D)^2/\pi(1-\pi)$ for squared error); and a prior sitting on a kink — the companion's symmetric case — is the polyhedral configuration yielding threshold-free linear response (§6). The dead zone is the sharp polyhedral form of the Radner–Stiglitz nonconcavity of the value of information [13, 14]: for asymmetric states under a polyhedral objective, weakly-correlated feedback is *worthless*, not merely weak, and the sign of the correlation matters. (**C5**) We order the three envelopes (mixing, information, bit-budget) below the master total-variation envelope and prove mixing and information are incomparable — for every lag and every K, a *preview process* keeps $\beta(D) \geq \frac{1}{2}$ while the information envelope, and the true value, vanish with K (§5). (**C6**) A nine-series battery across four physical domains bounds any undetected history excess within a registered equivalence margin where surrogates predict none, observes the envelope's tightness, and — read backwards — uses the incremental theorem as a certified witness of *conditional* history information on non-Markov series, always through a one-sided model-calibrated lower bound: $I(X_t; \text{history} \mid X_{t-D}) \geq 2[(\widehat{\Delta}_W - \widehat{\Delta}_Z - \varepsilon)^+]^2$ (§7).

The scope is stated precisely: this is a unified theory for *exogenous one-step bounded-reward decisions* under delay — the action does not move the state, the reward is single-stage. Within that scope the theory is closed on itself: one decision-theoretic object explaining an exact systems law, envelopes ordered by what they require (a model, samples, or a bit rate), sharpness results tying envelope to law, impossibility results delimiting both, and a falsifiable empirical program run on real processes from four physical domains.

2 Setting and the exact object

Let X be the current exogenous state taking values in a finite set $\mathcal{X}$, with marginal $\bar{\mu}$. A controller observes a delayed quantity Y (a delayed sample X_{t-D}, a delayed history, or any $\sigma(\mathcal{F}_{\leq t-D})$-measurable statistic), chooses an action w, and receives a bounded one-step reward $R(w,x)$ with oscillation $\Omega = \max_{w,x} R - \min_{w,x} R$. Define the value of a belief $\mu \in \Delta(\mathcal{X})$:

$$\psi(\mu) \;=\; \max_{w} \sum_{x} R(w,x)\,\mu(x),$$

a pointwise maximum of linear functionals, hence convex; and the posterior $\mu_Y = \mathbb{P}(X \in \cdot \mid Y)$, with $\mathbb{E}[\mu_Y] = \bar{\mu}$ by the tower property. The optimal Y-measurable controller attains $V_Y^\star = \mathbb{E}[\psi(\mu_Y)]$; the best static controller attains $V_{\text{stat}} = \psi(\bar{\mu})$. The exact object of this paper is the **Bayes envelope**

$$\Delta_Y \;=\; V_Y^\star - V_{\text{stat}} \;=\; \mathbb{E}\big[\psi(\mu_Y)\big] - \psi(\bar{\mu}) \;\geq\; 0, \tag{1}$$

nonnegative by Jensen. Everything that follows is an envelope on (1); the companion paper's exact law is the case where (1) admits a closed form.

Lemma 1 (Lipschitz value [1, App. A])**.** $|\psi(\mu)-\psi(\mu')| \leq \Omega\,\|\mu-\mu'\|_{\text{TV}}$, *hence* $\Delta_Y \leq \Omega\,\mathbb{E}\|\mu_Y-\bar{\mu}\|_{\text{TV}}$.

The total-variation route specializes to the β-mixing converse of [1] when Y is the full delayed history: $\Delta(D) \leq \Omega\,\beta(D)$.

3 The predictive-information envelope

Theorem 2 (Predictive-information bound)**.** *For any bounded objective with oscillation Ω and any delayed observable Y,*

$$0 \;\leq\; \Delta_Y \;\leq\; \Omega\sqrt{\tfrac{1}{2}\,I(X;Y)} \qquad (I \textit{ in nats; in bits, } \Omega\sqrt{\tfrac{\ln 2}{2}\,I_{\text{bits}}}). \tag{2}$$

Proof. By Lemma 1, $\Delta_Y \leq \Omega\,\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\text{TV}}$. Pinsker's inequality gives, for every realization y, $\|\mu_y - \bar{\mu}\|_{\text{TV}} \leq \sqrt{\tfrac{1}{2}D_{\text{KL}}(\mu_y\|\bar{\mu})}$. Taking the expectation over Y and applying Jensen to the concave square root,

$$\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\text{TV}} \;\leq\; \mathbb{E}\sqrt{\tfrac{1}{2}D_{\text{KL}}(\mu_Y\|\bar{\mu})} \;\leq\; \sqrt{\tfrac{1}{2}\,\mathbb{E}\,D_{\text{KL}}(\mu_Y\|\bar{\mu})} \;=\; \sqrt{\tfrac{1}{2}\,I(X;Y)},$$

the last equality being the definition of mutual information as the expected KL divergence of posterior from prior. □

Corollary 3 (Bit-limited telemetry)**.** *Suppose the delivered observation satisfies $H(Y) \leq b\ln 2$ nats (in particular, whenever Y takes at most 2^b values, or is the output of a prefix code of expected length $\leq b$ bits). Then*

$$\Delta_Y \;\leq\; \Omega\,\min\Big\{1,\;\sqrt{(b\ln 2)/2}\Big\},$$

the clip because the trivial range bound $\Delta_Y \leq \Omega$ beats the square root already at $b \geq 3$ bits.

Lemma 4 (Garbling never helps: value monotonicity)**.** *If $Y = g(Z)$ (deterministically or through any channel acting on Z alone), then $\Delta_Y \le \Delta_Z$, and also $\Delta_Y \le \Omega\sqrt{\frac{1}{2}I(X;Z)}$.*

Proof. $\mu_Y = \mathbb{E}[\mu_Z \mid Y]$ by the tower property, so convexity of ψ and Jensen give $\mathbb{E}\psi(\mu_Y) \le \mathbb{E}\psi(\mu_Z)$; subtract $\psi(\bar{\mu})$. The second claim is Theorem 2 plus data processing ($I(X;Y) \le I(X;Z)$). Note the two statements are different in kind: data processing alone bounds only the information *envelope*; the Jensen step bounds the *value itself* — the Blackwell comparison-of- experiments order [12] specialized to garbled delayed sensing. Together: compression can preserve feedback value; no encoding of the delayed signal can create it. □

The same tower-property mechanism yields the theorem that governs *richer* observations — the form §7 actually needs:

Theorem 5 (Incremental-information envelope)**.** *Let Z be any delayed observable and $W = (Z, H)$ a refinement (e.g. $Z = X_{t-D}$ a single delayed sample, H the rest of the delayed history window). Then*

$$0 \;\le\; \Delta_W - \Delta_Z \;\le\; \Omega\sqrt{\tfrac{1}{2}\, I(X; H \mid Z)}.$$

Equivalently, read backwards: incremental decision value certifies conditional predictive information*,*

$$I(X; H \mid Z) \;\ge\; 2\Big(\frac{\Delta_W - \Delta_Z}{\Omega}\Big)^2.$$

Proof. $\mu_Z = \mathbb{E}[\mu_W \mid Z]$ by the tower property, so $\Delta_W - \Delta_Z = \mathbb{E}[\psi(\mu_W) - \psi(\mu_Z)] \ge 0$ by Jensen (Lemma 4), and pointwise $\psi(\mu_W) - \psi(\mu_Z) \le \Omega\|\mu_W - \mu_Z\|_{\mathrm{TV}}$ (Lemma 1). Pinsker and Jensen then give $\mathbb{E}\|\mu_W - \mu_Z\|_{\mathrm{TV}} \le \sqrt{\frac{1}{2}\,\mathbb{E}\, D_{\mathrm{KL}}(\mu_W \| \mu_Z)} = \sqrt{\frac{1}{2}\, I(X; H \mid Z)}$, the last equality being the definition of conditional mutual information as the expected KL divergence of the refined posterior from the coarse one. □

Theorem 2 is the case $Z = \varnothing$. The pair separates two questions a delayed sensor can ask: *how much is the sample worth?* (Theorem 2 at $Y = Z$) and *how much more is the history worth?* (Theorem 5) — and the second is bounded by information the single sample does *not* carry, never by the single-sample information itself.

Proposition 6 (Leading-order tightness in the canonical case)**.** *For a stationary fair binary pair with lag-D correlation $r = r(D)$,*

$$I(X_t; X_{t-D}) = \tfrac{1}{2}\big[(1{+}r)\ln(1{+}r) + (1{-}r)\ln(1{-}r)\big] = \tfrac{r^2}{2} + \tfrac{r^4}{12} + O(r^6),$$

an even function of r, and the exact single-sample value of the $0/1$ prediction objective is $\Delta(D) = \frac{1}{2}|r(D)|$ (negative correlation is exploited by reversing the prediction). The right side of (2) equals $\Omega\big(\frac{|r|}{2} + O(|r|^3)\big)$: the envelope matches the exact law to leading order at $\Omega = 1$. (The companion's persistent-chain assumption — flip probability $\le \frac{1}{2}$ — gives $r \ge 0$ and its statement $\frac{1}{2}r$; we carry the absolute value throughout.) The information envelope is not loose where it matters.

Proof. The joint distribution places mass $\frac{1+r}{4}$ on each agreeing pair and $\frac{1-r}{4}$ on each disagreeing pair; the stated I follows by direct computation, and the expansion from $(1 \pm r)\ln(1 \pm r) = \pm r + \frac{r^2}{2} \mp \frac{r^3}{6} + \cdots$. Then $\sqrt{\frac{1}{2}I} = \frac{r}{2}\sqrt{1 + \frac{r^2}{6} + O(r^4)}$. □

4 No universal equality $\Delta = f(I)$

Proposition 7 (Information is a scalar; value is a geometry)**.** *There is a state space, a bounded objective, and two observations Y_1, Y_2 with $I(X;Y_1) = I(X;Y_2) > 0$ but $\Delta_{Y_1} = \frac{\Omega}{2} > 0 = \Delta_{Y_2}$. Hence no function f with $\Delta_Y = f(I(X;Y))$ exists for all experiments.*

Proof. Let $X = (B_1, B_2)$ be two independent fair bits and let the objective depend only on B_1: actions $w \in \{0,1\}$, $R(w,x) = \Omega\,\mathbf{1}[w = B_1]$ (so the oscillation is Ω). Static value: $\Omega/2$. Let $Y_1 = B_1$ and $Y_2 = B_2$. Both have $I(X;Y_i) = \ln 2$. Observing Y_1 yields $V^\star = \Omega$, so $\Delta_{Y_1} = \Omega/2$; observing Y_2 leaves the posterior over B_1 uniform, so $\Delta_{Y_2} = 0$. The two experiments share every information quantity in $I(X;Y)$ yet differ maximally in value: the envelope (2) is attained within a constant on one and vacuous on the other. $\square$

Remark 8. *This is the precise sense in which the companion paper's binary law is special: there the state is one decision-relevant bit, so the geometry collapses to a one-parameter family and the envelope inverts to an exact closed form. The general exact object remains (1).*

5 Envelope comparison

The three envelopes are not interchangeable, and their relationship is an ordered pair of relaxations of one master quantity.

Proposition 9 (Master envelope and universal ordering)**.** *For any delayed observable Y measurable w.r.t. the delayed history $\mathcal{F}_{\le t-D}$,*

$$\Delta_Y \;\le\; \Omega\,\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}} \;\le\; \Omega\,\min\Big\{\beta(D),\ \sqrt{\tfrac{1}{2}I(X;Y)}\Big\},$$

where $\beta(D)$ is the β-mixing coefficient of the state process at lag D. The expected posterior movement $\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}}$ is the master envelope; the mixing and information envelopes are its two relaxations.

Proof. The first inequality is Lemma 1. For the second: since Y is $\mathcal{F}_{\le t-D}$-measurable and $\sigma(X) \subseteq \mathcal{F}_{\ge t}$, conditioning on the coarser Y can only shrink expected posterior movement, and $\mathbb{E}\|\mu_{\mathcal{F}_{\le t-D}} - \bar{\mu}\|_{\mathrm{TV}} \le \beta(D)$ because the β-coefficient takes its supremum over the larger future field [1, App. A]. The information relaxation is the Pinsker step in the proof of Theorem 2. $\square$

Throughout, we use the companion's event-supremum convention

$$\beta(D) \;=\; \mathbb{E}\Big[\sup_{B \in \mathcal{F}_{\ge t}} \big|\mathbb{P}(B \mid \mathcal{F}_{\le t-D}) - \mathbb{P}(B)\big|\Big],$$

which lower-bounds the absolute-regularity coefficient; conventions differ by factors of two across the literature, and every displayed constant below is tied to this one. Lower bounds on this β are what vacuousness of the mixing envelope requires.

Proposition 10 (The two relaxations are incomparable)**.** *Neither relaxation dominates the other.*

(i) Mixing strictly tighter: *for the symmetric two-state chain with $Y = X_{t-D}$, $\beta(D) = \frac{1}{2}|r|$ while $\sqrt{\frac{1}{2}I} = \frac{|r|}{2}\sqrt{1 + \frac{r^2}{6} + O(r^4)} > \beta(D)$ for every $|r| \in (0,1)$.*

(ii) Information arbitrarily tighter: *for every fixed lag D and every $K \geq 2$ there is a stationary state process and bounded objective with*

$$\beta(D) \;\geq\; \tfrac{1}{2}, \qquad I(X_t; \mathcal{F}_{\leq t-D}) \;\leq\; h_2(1/K) + \tfrac{\ln 2}{K} \;\xrightarrow[K\to\infty]{}\; 0, \qquad \Delta \;=\; \tfrac{\Omega}{2K},$$

where h_2 is the binary entropy in nats: the mixing envelope stays $\geq \Omega/2$ while the information envelope, and the true value, vanish.

Proof. (i) is Proposition 6 together with $\beta(D) = \frac{1}{2}|r(D)|$ for the binary chain (Appendix A). (ii) *The preview process (one per (D, K)).* Fix D and K, set $m = D + K$. Let c_t be i.i.d. fair bits, let $\Theta \sim \mathrm{Unif}\{0, \ldots, K-1\}$ be an independent phase (randomized to make the pair stationary), and let the observable state be $X_t = (c_t, p_t)$ with

$$p_t \;=\; \begin{cases} c_{t+m} & t \equiv \Theta \pmod K \\ \perp & \text{otherwise,} \end{cases}$$

a sensor that occasionally publishes a *forecast* m steps ahead. The objective depends only on the current payoff bit: $R(w, x) = \Omega\,\mathbf{1}[w = c]$. *Mixing (lower bound).* The window $[t-m, t-D-1]$ contains K consecutive integers, hence exactly one preview slot $s \equiv \Theta$; that preview is in the delayed past and reveals c_{s+m} with $s + m \in [t, t + K]$, a present-or-future payoff bit. The event $B = \{c_{s+m} = 1\} \in \mathcal{F}_{\geq t}$ then has conditional probability in $\{0, 1\}$ against marginal $\frac{1}{2}$, so $\beta(D) \geq \frac{1}{2}$. (Previews shared between past and future create further dependence; we claim only the lower bound, which is what vacuousness of the mixing envelope needs.) *Information (upper bound).* X_t is determined by the triple (payoff bit c_t; slot indicator $\mathbf{1}[t \equiv \Theta]$; preview value when present, a *future* bit independent of the past). The delayed history reveals the slot indicator (it determines Θ) and reveals c_t itself exactly when $t - m \equiv \Theta$ (probability $1/K$); hence $I(X_t; \mathcal{F}_{\leq t-D}) \leq h_2(1/K) + \frac{\ln 2}{K}$. *Value.* Feedback wins $\Omega/2$ on the revealed fraction $1/K$ and nothing elsewhere: $\Delta = \Omega/(2K)$. The information envelope $\Omega\sqrt{(h_2(1/K) + \ln 2/K)/2}$ decays with K (rate $\widetilde{O}(K^{-1/2})$ against the true K^{-1} — loose, but not blind); the mixing envelope does not decay at all. $\square$

Remark 11 (Why keep the information form at all)**.** *Proposition 9 shows the TV master envelope is always at least as tight, and (i) shows mixing can beat information. Three properties are the information envelope's own: it is* estimable *— the plug-in $\hat{I}(X_t; X_{t-D})$ in §7 needs only pair counts, while the β-coefficient's supremum over the future σ-field is not estimable without a process model; it* carries rate semantics *— total-variation dependence coefficients also contract under channels, but only information obeys the entropy bound $I(X; Y) \leq H(Y)$ that turns a bit budget into Corollary 3; and it obeys an* exact conditional chain rule*, giving additive multi-sensor budget decompositions via conditional mutual information (Theorem 5) that β does not. In short: mixing is the sharper analysis tool when a model is in hand; information is the operational one when only data or a bit budget is.*

6 Exactness: a complete account for binary state, and its limits

When does the Bayes envelope admit a closed form — an exact *law* rather than an envelope? For binary state with a finite action set, the exact law is a sum of hinges; for general convex value functions, the *local contact order* of the value function at the prior determines the weak-correlation exponent (Theorem 14).

Envelope	Bound on Δ_Y	Needs	Blind spot
master (TV)	$\Omega\,\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}}$	posterior process	not estimable directly
mixing	$\Omega\,\beta(D)$	process model	future events beyond X_t (Prop. 10ii)
information	$\Omega\sqrt{I(X;Y)/2}$	pair samples	decision geometry (Prop. 7)
bit budget	$\Omega\sqrt{(b\ln 2)/2}$	channel rate only	everything but the rate

Table 1: The envelope family. The structure is a *branching*, not a hierarchy: both mixing and information relax the master TV envelope, in incomparable directions (Proposition 10), and the bit budget further relaxes the information branch via Lemma 4: TV $\to$ { mixing; information $\to$ bit budget }. Each branch trades knowledge for tightness in its own currency; the battery of §7 measures the first three on real processes.

Theorem 12 (The hinge law, polyhedral case)**.** *Let $X_t \in \{0,1\}$ be stationary with $\pi = \mathbb{P}(X_t{=}1) \in (0,1)$, let $Y = X_{t-D}$, and write $C = C(D) = \mathrm{Cov}(X_t, X_{t-D})$. Assume the action set is finite — or, more generally, that the value function $\psi(p) = \sup_w[pR(w,1) + (1-p)R(w,0)]$ is* polyhedral *(a maximum of finitely many affine functions of the belief), with kinks $p_1^* < \cdots < p_K^*$ and slope jumps $\Delta s_k > 0$. Then, exactly,*

$$\Delta(C) \;=\; \begin{cases} \displaystyle\sum_k \Delta s_k\,\big(C - a_k^+\big)^+, & C \ge 0, \\ \displaystyle\sum_k \Delta s_k\,\big((-C) - a_k^-\big)^+, & C < 0, \end{cases} \qquad a_k^+ = \begin{cases} (p_k^* - \pi)\,\pi, & p_k^* \ge \pi, \\ (\pi - p_k^*)(1-\pi), & p_k^* < \pi, \end{cases} \qquad a_k^- = \begin{cases} (p_k^* - \pi)(1-\pi), & p_k^* \ge \pi, \\ (\pi - p_k^*)\,\pi, & p_k^* < \pi. \end{cases}$$

Δ is piecewise-linear and nondecreasing in $|C|$ on each covariance-sign branch*, identically zero on the branch dead zone; unless $\pi = \frac{1}{2}$ or the value function has the corresponding reflection symmetry about π (kink locations* and *slope jumps), the two branches generally differ — positive and negative correlation of the same magnitude can have different value, one of them zero.*

Proof. Write $\psi(p) = \psi(p_1^*) + s_0(p - p_1^*) + \sum_k \Delta s_k (p - p_k^*)^+$ (polyhedral by hypothesis). Observing Y produces the two-point posterior $p_1 = \pi + C/\pi$ (w.p. π) and $p_0 = \pi - C/(1-\pi)$ (w.p. $1-\pi$), whose mean is π; the affine part of ψ cancels in $\Delta = \mathbb{E}[\psi(p_Y)] - \psi(\pi)$, leaving $\Delta = \sum_k \Delta s_k\{\mathbb{E}(p_Y - p_k^*)^+ - (\pi - p_k^*)^+\}$. Fix k. For $C \ge 0$ ($p_0 \le \pi \le p_1$): if $p_k^* \ge \pi$, $\mathbb{E}(p_Y - p_k^*)^+ = \pi(p_1 - p_k^*)^+ = (C - (p_k^* - \pi)\pi)^+$ and $(\pi - p_k^*)^+ = 0$; if $p_k^* < \pi$, the pointwise identity $(x)^+ - x = (-x)^+$ gives $\mathbb{E}(p_Y - p_k^*)^+ - (\pi - p_k^*) = \mathbb{E}(p_k^* - p_Y)^+ = (1-\pi)(p_k^* - p_0)^+ = (C - (\pi - p_k^*)(1-\pi))^+$. For $C < 0$ the posteriors trade places ($p_1 < \pi < p_0$): if $p_k^* \ge \pi$, only p_0 can exceed the kink and $\mathbb{E}(p_Y - p_k^*)^+ = (1-\pi)(p_0 - p_k^*)^+ = ((-C) - (p_k^* - \pi)(1-\pi))^+$; if $p_k^* < \pi$, the same identity gives $\pi(p_k^* - p_1)^+ = ((-C) - (\pi - p_k^*)\pi)^+$. Summing over k gives the two branches. $\square$

6.1 Beyond polyhedral objectives: contact-order response

The polyhedral hypothesis is not decoration. For a continuous action set ψ is a supremum of *infinitely* many affine functions and need not be piecewise-linear, and the hinge law fails — constructively:

Proposition 13 (Squared-error prediction: exactly quadratic)**.** *For $w \in [0,1]$ and $R(w,x) = -(w-x)^2$ (oscillation $\Omega = 1$), $\psi(p) = -p(1-p)$ is smooth and strictly convex, and, exactly,*

$$\Delta(D) \;=\; \mathrm{Var}(p_Y) \;=\; \frac{C(D)^2}{\pi(1-\pi)}\;:$$

quadratic in the covariance, no hinge, no dead zone, for every π. More generally, whenever ψ is twice continuously differentiable near π with $\psi''(\pi) > 0$ (if the curvature vanishes the response is of higher order; Theorem 14),

$$\Delta(D) \;=\; \tfrac{1}{2}\,\psi''(\pi)\,\mathrm{Var}(p_Y) + o(C^2) \;=\; \frac{\psi''(\pi)}{2\pi(1-\pi)}\,C(D)^2 + o\big(C(D)^2\big).$$

Proof. The optimal action under belief p is $w = p$, so $\psi(p) = -p(1-p) = p^2 - p$; then $\Delta = \mathbb{E}[\psi(p_Y)] - \psi(\pi) = \mathbb{E}[p_Y^2] - \pi^2 = \mathrm{Var}(p_Y)$, exact because ψ is quadratic, and $\mathrm{Var}(p_Y) = \pi(C/\pi)^2 + (1-\pi)\big(C/(1-\pi)\big)^2 = C^2/\pi(1-\pi)$. The general statement is the second-order Taylor expansion of ψ at π against the mean-π posterior spread. □

Both regimes — and every intermediate one — are cases of a single statement:

Theorem 14 (Local response is set by the contact order)**.** *Suppose that near π,*

$$\psi(\pi+h) \;=\; \psi(\pi) + ah + c_+\,h_+^q + c_-\,(-h)_+^q + o(|h|^q)$$

for some $q \geq 1$ and $c_+, c_- \geq 0$ not both zero. Then, as $C \downarrow 0$,

$$\Delta(C) \;=\; C^{\,q}\Big[c_+\,\pi^{1-q} + c_-\,(1-\pi)^{1-q}\Big] + o(C^{\,q}),$$

and on the negative branch ($C \uparrow 0$) the same with $|C|^q$ and the weights $\pi \leftrightarrow (1-\pi)$ interchanged. If instead ψ is affine on a neighbourhood of π, then $\Delta(C) = 0$ for all $|C|$ small enough that both posteriors remain in that neighbourhood — the exact dead zone.

Proof. The posteriors are $p_1 = \pi + C/\pi$ (w.p. π) and $p_0 = \pi - C/(1-\pi)$ (w.p. $1-\pi$); their mean is π, so the affine part of the expansion cancels in $\Delta = \mathbb{E}[\psi(p_Y)] - \psi(\pi)$. For $C > 0$ the h_+^q term contributes $\pi\,c_+(C/\pi)^q = c_+C^q\pi^{1-q}$ through p_1 and the $(-h)_+^q$ term contributes $(1-\pi)\,c_-(C/(1-\pi))^q = c_-C^q(1-\pi)^{1-q}$ through p_0; both posterior deviations are $\Theta(C)$, so the $o(|h|^q)$ remainder is $o(C^q)$. For $C < 0$ the posteriors trade sides of π, exchanging the weights. The affine case is immediate. □

Remark 15 (The trichotomy, as leading cases)**.** *Theorem 14 subsumes the named regimes — $q = 1$ (kink): linear response; locally affine: dead zone; $q = 2$ with $c_\pm = \frac{1}{2}\psi''(\pi) > 0$: quadratic — and admits fractional contact orders $q \in (1,2)$, realizable by valid suprema of affine rewards, with correspondingly fractional correlation exponents. Each case fixes how weak dependence buys value:* prior on a kink $\Rightarrow$ *linear response $\Delta \propto |C|$;* prior interior to a flat piece $\Rightarrow$ *dead zone, $\Delta = 0$ until a threshold;* smooth curvature $\Rightarrow$ *quadratic response $\Delta \propto C^2$. The companion's law is the first case — symmetry puts the prior on the decision kink, the polyhedral configuration yielding threshold-free linear response. The dead zone is the sharp polyhedral form of the Radner–Stiglitz nonconcavity of the value of information [13, 14], whose smooth setting is our third case (zero marginal value at zero information, but not identically zero). And the sign-dependence of Theorem 12 means even* the direction *of correlation matters away from symmetry: with $\pi = 0.2$ and one kink at $p^* = 0.3$, covariance* $+0.03$ *clears its threshold* $(0.1)(0.2) = 0.02$ *and has value, while* -0.03 *sits inside its threshold* $(0.1)(0.8) = 0.08$ *and is worthless.*

Corollary 16 (The companion law, and why it is threshold-free)**.** *For the symmetric chain ($\pi = \frac{1}{2}$) with the 0/1 prediction objective (K=1, $p^* = \frac{1}{2}$, $\Delta s = 2$): $C^* = 0$ and $|C(D)| = \frac{1}{4}|r(D)|$, so $\Delta(D) = 2 \cdot \frac{1}{4}|r(D)| = \frac{1}{2}|r(D)|$ — the exact law of [1]. Symmetry places the prior* on *the kink — the polyhedral configuration with no dead zone; the linear-in-r law is the degenerate hinge.*

Remark 17 (The dead zone is a prediction, not a curiosity)**.** *For $\pi \neq p^*$ the hinge says weakly-correlated delayed feedback is* worthless *— not slightly valuable — until $|C(D)|$ crosses C^*: below threshold, both posteriors fall on one side of the decision kink and the optimal action never changes. Operationally: a rare-event admission controller (π small, p^* moderate) gains* nothing *from telemetry until the covariance clears the* branch *threshold: $C(D) > (p^* - \pi)\pi$ for positive correlation, but $-C(D) > (p^* - \pi)(1 - \pi)$ for negative — for rare events (π small) the negative-branch threshold is larger by the factor $(1 - \pi)/\pi$, so anticorrelated telemetry must be far stronger before it is worth anything — an exact, testable, sign-dependent cutoff for when to stop paying for freshness at all. None of the envelopes of §5 see this: the dead zone is a fact about geometry (posteriors vs. kink), which is precisely what scalar information summaries discard (Proposition 7).*

Remark 18 (Beyond two states: the layer-cake lift, and the boundary)**.** *The companion's layer-cake result [1, Prop. 2] lifts binary exactness to capacities $a_t \in \{0, \dots, C\}$ with monotone threshold objectives: the objective decomposes over layer indicators $x_t^u = \mathbf{1}[a_t \geq u]$ and the total advantage is the exact sum $\Delta(D) = \sum_u \pi_u r_u(D)$ — each layer a binary experiment, each governed by its hinge (there with symmetric thresholds, so zero dead zones). Effective one-dimensionality of the posterior family (a scalar belief per layer, moved along a line by the observation) is a principal tractable regime in which scalar closed forms arise; the two-bit construction of Proposition 7 is the minimal experiment whose posterior geometry is genuinely two-dimensional, and there the value ceases to be a function of any scalar summary of dependence — correlation or information alike.*

7 The law and its envelopes across domains

The companion paper had room for one paragraph of this study; the full protocol and results are reported here. The question has two halves. First, *single-sample sufficiency*: on real binary state processes, does history beyond the single delayed sample add decision value beyond the *exact empirical single-sample Bayes value* — computed from the pooled 2×2 lag table with no symmetry assumption, so that Theorem 12's imbalance effects (dead zones included) are carried automatically? Second, *the envelope*: does the measured value respect, and how tightly does it approach, the predictive-information bound of Theorem 2?

7.1 Protocol

States. Binary regime series from four physical domains plus controls: acoustic source azimuth and voice activity (LOCATA, 120 Hz); sperm-whale coda type (Dominica corpus, per-exchange); GOES space-weather series (X-ray flux, magnetometer, proton flux; 1–5-minute cadence); seismic station regime (ANMO, 1-minute); a synthetic two-state Markov chain (flip probability 0.02) as positive reference; an i.i.d. fair-bit series as null. Each series is binarized by the domain's registered thresholding (the companion study's preparation scripts, unchanged).

The two estimands. For each lag D on a per-domain grid: (i) the *exact empirical single-sample Bayes value* $\widehat{V}_1(D) = \sum_y \hat{w}_y \max(\hat{p}_y, 1-\hat{p}_y) - \max(\hat{\pi}, 1-\hat{\pi})$, the plug-in of (1) for $Y = X_{t-D}$ computed from the pooled lag table — for a balanced series this is $\frac{1}{2}|r(D)|$, and for an imbalanced one it carries Theorem 12's thresholds automatically (the per-domain class proportion $\hat{\pi}$ is reported alongside); (ii) the value of the best *delayed-history* policy: a k=6-lag lookup predictor trained and evaluated by blocked cross-validation (5 folds, contiguous blocks), reported as accuracy above the *best static (majority) policy* $\max(\hat{\pi}, 1 - \hat{\pi})$ — the correct static benchmark; accuracy above $\frac{1}{2}$ overstates value whenever $\hat{\pi} \neq \frac{1}{2}$. The *history gain* is the difference (ii)$-$(i).

Scope of the estimand. Estimand (ii) is the value of one *registered policy class* $\mathcal{P}$ (six-lag lookup), i.e. $\widehat{\Delta}_{\mathcal{P},W} = \sup_{\pi\in\mathcal{P}} V(\pi; W) - V_{\text{stat}}$, *not* the Bayes history value Δ_W of Theorem 5. The asymmetry matters. Since $V(\pi; W) \le V_W^\star$ for every π, any demonstrated improvement over the exact single-sample value lower-bounds $\Delta_W - \Delta_Z$ — so *positive* certificates transfer upward and the conditional-information witnesses below are sound. *Negative* findings bound only the class: where the lookup policy finds nothing, some richer policy class could still extract history value, so verdicts of that kind are stated as "no additional value detected by the registered six-lag lookup policy," never as unrestricted single-sample sufficiency.

Surrogate calibration. The null for "history adds nothing beyond the law" is a population of Markov surrogates per domain, each matched to the real series' lag-1 transition probabilities and evaluated by the identical pipeline, with $M = 299$ surrogates. We name the procedure precisely: because the surrogates are generated from *estimated* lag-1 transition probabilities, this is a *parametric-bootstrap test under a fitted Markov null* — not an exact randomization test for the broad null "history adds nothing" — with $p = (1 + \#\{T_j \ge T_{\text{obs}}\})/(M + 1)$ (smallest reportable value $1/300 \approx 0.0033$, written $p < .004$, never .00), together with the companion study's practical floor of 0.03 on the effect. Because each surrogate is Markov, its true history gain is zero and its own exact $\widehat{V}_1$ is computable from its lag table, so the M surrogate gain estimates sample the pipeline's estimation error E *under the fitted Markov model.* The two tails play different roles: a lower certificate needs the *over*-estimation tail, $\varepsilon_{\text{over}} = q_{0.95}(E)$, giving $L = \hat{g} - \varepsilon_{\text{over}}$; an upper (equivalence) bound needs the *under*-estimation tail, $\varepsilon_{\text{under}} = -q_{0.05}(E)$, giving $U_2 = \hat{g} + \varepsilon_{\text{under}}$; the two coincide only under a symmetry condition, so both are computed and reported separately. Both are *model-calibrated* (not distribution-free) bounds. Complementing them, the analysis reports a *dependence-aware* moving-block bootstrap on held-out paired loss differences — the registered k=6 policy against the pooled-*exact* single-sample policy (in-sample fit favouring the single-sample arm, hence conservative for the incremental value), blocks resampled within runs at the selected lag D^*. Because D^* is an argmax over the lag grid that the bootstrap treats as fixed, the i.i.d. null's paired interval calibrates the *selection* bias not covered (its upper end ≈ 0.02); paired lower bounds are claimed only above that floor, and repeating the lag selection inside each bootstrap replicate is the registered refinement for the camera-ready analysis. Validity checks pass: the Markov reference's interval straddles zero (no phantom conditional information) and azimuth's is slightly negative (the overfitting cost of the richer class). *Non-rejection is not exactness*: where the test does not fire, we report the model-calibrated upper bound $U = \text{gain}_{\max} + \varepsilon_{\text{dom}}$ and claim "no detectable excess (equivalent within the margin)" only when $U < 0.03$.

The two information measurements. For each domain and lag, the plug-in $\widehat{I}(X_t; X_{t-D})$ from pooled pair counts (2×2; plug-in bias $\le 3/(2N)$ nats, negligible at the pooled sample sizes reported) and the envelope $\sqrt{\widehat{I}/2}$. Their registered uses follow the two theorems, matched to the observable each governs. *Single sample* (Theorem 2 at $Y = X_{t-D}$): the exact $\widehat{V}_1$ must respect $\sqrt{\widehat{I}/2}$, a population identity whose empirical check is a consistency diagnostic. *History* (Theorem 5): the history policy observes the *window*, so the single-lag envelope does not bound it — a history value above $\sqrt{\widehat{I}_1/2}$ is not a violation but exactly what genuine higher-order dependence should produce. The history gain over $\widehat{V}_1$ is the empirical increment $\Delta_W - \Delta_Z$, and its model-calibrated lower bound certifies *conditional* predictive information:

$$I(X_t;\, \text{history} \mid X_{t-D}) \;\ge\; 2\left[(\text{gain}_{\max} - \varepsilon_{\text{dom}})^+\right]^2,$$

never through the raw point estimate, whose positive noise would inflate the certificate. The total-information witness $I(X_t; \text{window}) \ge 2[(\widehat{\Delta}_{\text{hist}} - \varepsilon_{\text{dom}})^+]^2$ is retained separately.

7.2 Results

Table 2 reports the final analysis (battery v5; `battery_v5.json`, `battery_v5b.json`; the v3/v4 development runs and their protocol corrections are preserved in the repository record). Per domain: the class proportion $\hat{\pi}$; the maximum history-class gain over the exact single-sample Bayes value $\widehat{V}_1$; the parametric-bootstrap p under the fitted Markov null (M=299, floor 1/300, written $< .004$); the two calibration tails $\varepsilon_{\text{over}} = q_{0.95}(E)$ and $\varepsilon_{\text{under}} = -q_{0.05}(E)$; the two-tail equivalence bound $U_2 = \text{gain}_{\max} + \varepsilon_{\text{under}}$ (margin 0.03); and the dependence-aware paired interval — a moving-block bootstrap (blocks within runs, B=2000) of the per-position loss difference between the registered k=6 policy and the pooled-*exact* single-sample policy at the selected lag D^*, the exact-arm fit favouring the single-sample side so the interval is conservative for the incremental value.

Domain	$\hat{\pi}$	gain	p	ε_{ov}	ε_{un}	U_2	paired 95% CI	verdict
Two-state Markov	.470	+.000	.78	.006	.000	.000	$[-.003, +.014]$	equivalent
i.i.d. null	.497	+.012	.023	.011	$-.001$	.011	$[+.005, +.020]$	selection floor
LOCATA azimuth	.494	+.008	.063	.009	.020	.027	$[-.003, -.000]$	equivalent
LOCATA VAD	.636	$-.000$	.82	.052	.000	.000	$[.000, .000]$	equivalent
Sperm-whale codas	.483	+.046	1.0	.117	$-.063$	$-.016$	$[-.055, -.018]$	equivalent
GOES protons	.489	+.029	.11	.033	.003	.032	$[+.002, +.051]$	not established
GOES X-ray flux	.421	+.028	$< .004$	.008	.001	.028	$[+.011, +.037]$	detected, below floor
GOES magnetometer	.456	+.033	$< .004$	.010	.001	.033	$[+.022, +.048]$	detected (exploratory)
Seismic ANMO	.508	+.158	$< .004$	.014	.000	.158	$[+\mathbf{.146}, +\mathbf{.168}]$	history adds

Table 2: Battery v5 (final analysis). Verdicts are statements about the *registered six-lag lookup class*: "equivalent" = $U_2 < 0.03$ under both-tail calibration; "not established" = no detection but $U_2 \geq 0.03$; "detected (exploratory)" = statistically detected at the practical floor with a paired bound at the selection floor. The i.i.d. null's *positive* paired interval is the empirical lag-selection bias (its upper end, ≈ 0.02, is the claim floor for paired bounds, since D^* is an argmax the bootstrap treats as fixed). Bold: the one interval that dwarfs every calibration concern.

Four readings, in increasing order of consequence.

The protocol corrections changed the conclusions, honestly. Under the original chance baseline, four series appeared to carry history value beyond the single sample (gains 0.100–0.367). Under the majority baseline and exact $\widehat{V}_1$: VAD's gain vanished entirely ($+0.100 \to -0.000$; $\hat{\pi} = .636$ — the whole effect was the majority-class head start), X-ray shrank $0.185 \to 0.028$, magnetometer $0.147 \to 0.033$, and only seismic survived at strength ($0.367 \to 0.158$). The two-tail calibration then *strengthened* the equivalence side: $\varepsilon_{\text{under}}$ is near zero or negative on most series, so VAD and the whale codas join the class-equivalent block. Sub-unity envelope ratios disappeared with the imbalance correction (balanced-domain tightness ratios 1.11–1.35, on the Pinsker side of unity as population theory requires).

Only seismic exhibits a clearly material history gain. X-ray and magnetometer show smaller statistically detected gains at or below the registered practical threshold; on the remaining series *no additional value was detected by the registered six-lag lookup policy*, with class equivalence established where $U_2 < 0.03$ and honestly withheld where it is not (protons, $U_2 = .032$). None of these negative findings asserts unrestricted single-sample sufficiency: a richer policy class could in principle extract value the lookup class does not (§ protocol, scope of the estimand).

An empirical dead zone. The imbalanced GOES series exhibit Theorem 12's geometry at the plug-in level: X-ray ($\hat{\pi} = .421$, kink at $\frac{1}{2}$, positive-branch threshold $C^* \approx 0.033$) has $\widehat{V}_1$ exactly zero across its moderate-covariance lags — the majority policy is unbeatable by any single delayed sample — while its plug-in information remains positive: *information without value*, the phenomenon of Proposition 7, on a physical series. (Population uncertainty for the dead-zone boundary itself is future work; we label this an *empirical* dead zone.)

The conditional lower bound stands — once at strength. By Theorem 5, seismic's history gain converts into a lower bound on conditional predictive information. The *primary* result is the dependence-aware one: the moving-block interval $[+.146, +.168]$ gives $I(X_t; \text{history} \mid X_{t-D}) \geq 2(0.146)^2 \approx 0.043$ nats, with the stated caveat that D^* was selected before bootstrapping (the i.i.d.-calibrated selection floor, 0.02, is dwarfed sevenfold). The fitted-Markov surrogate route corroborates with $2(0.158 - 0.014)^2 \approx 0.041$ nats — two complementary calibration routes, agreeing to within a few percent — alongside the total-information witness $I(X_t; \text{window}) \geq 0.176$ nats. History knows what the last sample does not; that is what "aftershock clustering" means in information units. Magnetometer's paired bound (+.022) sits at the selection floor and is reported as exploratory. The instrument reads in both directions: class equivalence or bounded excess on most series; a dependence-aware conditional-information lower bound exactly where the history structure is found.

8 Related work

Value of information. The Bayes envelope is the classical object of statistical decision theory [2, 4], and its convex geometry is charted territory: De Lara and Gossner [3] study the value function as a support function, characterize when information has positive value, and give local estimates near uninformative experiments. Our Lipschitz lemma instantiates their payoffs–beliefs duality; the hinge and dead-zone analysis of §6 is the exact delayed-binary specialization of that convex-analytic programme; the comparison-of-experiments order behind Lemma 4 is Blackwell's [12]; and the Radner–Stiglitz nonconcavity [13, 14] anticipates the zero-marginal-value onset in smooth settings; the exact dead zone is its polyhedral counterpart. The information envelope itself is assembled from textbook inequalities, and its closest classical relatives are the Fano-type families bounding Bayes *error* by information; kindred moves appear wherever bounded value must be controlled by divergence, e.g. the information-ratio regret analyses of Russo and Van Roy [11]. To our knowledge the bounded-objective envelope (2) has not been stated as such; we accordingly claim assembly and anchoring, not the inequality. What we have not found elsewhere is the assembly this paper is about: the envelope *anchored to an exact law* (sharp to $O(r^3)$, with the $O(r^2)$ correction observed on real data), its ordering against mixing envelopes with both separations constructed, and its inversion into a certified information witness.

The logarithmic exception. For log-utility gambling the value of side information *equals* mutual information: Kelly [5] for the growth rate, Barron and Cover [6] for the general bound $\Delta \leq I$ with equality characterizations. Log utility is unbounded, escaping our Ω-bounded class; the pair of results brackets the landscape — unbounded log utility buys equality, bounded utility only an envelope, and Proposition 7 shows the gap is essential, not technical.

Age of information, and its critics. AoI [7] treats freshness as the object, with sampling-policy results such as [8] optimizing age processes. That plain age is not decision significance is already argued *within* the field: age-of-incorrect-information replaces the clock with an error-aware penalty [16], explicit AoI-vs-VoI scheduling comparisons exist for networked control [17], and value-of-information metrics for feedback control have been quantified directly, including an explicit derivation of the value–age relation [18, 15]. Our claim is accordingly not that age lacks decision semantics — that point is established — but sharper: for exogenous one-step bounded-reward decisions the exact semantic object is the posterior Bayes envelope, which admits a universal mutual-information converse, an explicit binary geometry (hinges, dead zones, quadratic regimes), and an operational inversion into predictive-information witnesses.

Delayed information in control. Delayed-sharing information structures go back to Witsenhausen [9] and are systematized by the common-information approach [10]; remote estimation studies the estimation face of the same trade [8]. Our setting is the tractable exogenous corner of that world — the action does not affect the state — which is exactly what makes closed forms and universal envelopes available.

Predictive information. Bialek, Nemenman and Tishby [19] study $I(\text{past}; \text{future})$ as a measure of process complexity. Theorems 2 and 12 are, in that vocabulary, statements about what predictive information can and cannot buy a bounded-utility decision maker — with the battery's witness bound $I \geq 2\widehat{\Delta}^2$ operating as a decision-theoretic estimator of predictive information from policy value alone.

9 Conclusion

The companion paper measured one exact law on one system. This paper locates that law inside a unified theory of exogenous one-step bounded-reward decisions under delay: value of delayed information is the Bayes envelope; the envelope is bounded by predictive information (universally), by mixing (given a model), and by the telemetry bit rate (given nothing); for binary state it is a sign-dependent sum of hinges under polyhedral objectives (smooth curvature giving quadratic weak-correlation response instead), exactly a layered sum for monotone multi-state objectives, and exactly nothing — inside the dead zone — for asymmetric weakly-coupled sensing. Beyond one effective dimension, no scalar summary of dependence determines value at all. Each claim is falsifiable, and the nine-series battery exercises all of them: no history excess is detectable where the equivalence bound requires none, the envelope's predicted tightness appears with its $O(r^2)$ correction, and history gains beyond the single-sample value certify conditional predictive information in nats. The instrument reads in both directions; that, more than any single inequality, is what we hope survives.

A Companion results restated

For self-containedness we restate the companion results used above, with proofs where short; [1] is under submission and unavailable to a reviewer.

Mixing coefficient. Throughout, $\beta(D) = \mathbb{E}[\sup_{B \in \mathcal{F}_{\geq t}} |\mathbb{P}(B \mid \mathcal{F}_{\leq t-D}) - \mathbb{P}(B)|]$ (event-supremum convention; a lower bound for the absolute-regularity coefficient, so the vacuousness results transfer conservatively).

Binary chain. For the symmetric two-state chain with flip probability p and $r(D) = (1-2p)^D$: conditioning on the delayed state moves the law of X_t by total variation $\frac{1}{2}|r(D)|$ from either value of the conditioner, and the Markov property collapses the past to the delayed sample; hence $\beta(D) = \frac{1}{2}|r(D)|$. The exact value of the 0/1 prediction objective is $\Delta(D) = \frac{1}{2}|r(D)|$: predict the delayed state if $r > 0$, its complement if $r < 0$; either beats the static $\frac{1}{2}$ by half the correlation magnitude.

Layer-cake lift (companion Prop. 2; statement and sketch). For capacity $a_t \in \{0, \dots, C\}$, threshold indicators $x_t^u = \mathbf{1}[a_t \geq u]$ with $\pi_u = \mathbb{P}(x_t^u{=}1)$, and objectives that are positive combinations of per-layer threshold rewards (goodput minus over-admission surplus in the companion's setting), the objective decomposes additively over layers via $\min(w,a) = \sum_u \mathbf{1}[w \geq u]\, x_t^u$ and $(w-a)^+ = \sum_u \mathbf{1}[w \geq u](1-x_t^u)$; the age-$D$ optimum applies the per-layer binary policy (the delayed indicators, whose nesting is inherited from the observation), and the total advantage over the best comonotone static budget at matched surplus is $\Delta(D) = \sum_u \pi_u r_u(D)$, $r_u(D) = \mathrm{corr}(x_t^u, x_{t-D}^u)$.

References

[1] A. H. Bond. Polite bursting: the value of delayed feedback in age-limited admission control. Under submission, IEEE INFOCOM 2027.

[2] R. A. Howard. Information value theory. *IEEE Trans. Systems Science and Cybernetics* 2(1):22–26, 1966.

[3] M. De Lara, O. Gossner. Payoffs-beliefs duality and the value of information. *SIAM J. Optimization* 30(1):287–310, 2020.

[4] H. Raiffa, R. Schlaifer. *Applied Statistical Decision Theory.* Harvard University, 1961.

[5] J. L. Kelly, Jr. A new interpretation of information rate. *Bell System Technical Journal* 35:917–926, 1956.

[6] A. R. Barron, T. M. Cover. A bound on the financial value of information. *IEEE Trans. Information Theory* 34(5):1097–1100, 1988.

[7] R. D. Yates, Y. Sun, D. R. Brown III, S. K. Kaul, E. Modiano, S. Ulukus. Age of information: an introduction and survey. *IEEE J. Selected Areas in Communications* 39(5):1183–1210, 2021.

[8] Y. Sun, E. Uysal-Biyikoglu, R. D. Yates, C. E. Koksal, N. B. Shroff. Update or wait: how to keep your data fresh. *IEEE Trans. Information Theory* 63(11):7492–7508, 2017.

[9] H. S. Witsenhausen. A counterexample in stochastic optimum control. *SIAM J. Control* 6(1):131–147, 1968.

[10] A. Nayyar, A. Mahajan, D. Teneketzis. Decentralized stochastic control with partial history sharing: a common information approach. *IEEE Trans. Automatic Control* 58(7):1644–1658, 2013.

[11] D. Russo, B. Van Roy. An information-theoretic analysis of Thompson sampling. *J. Machine Learning Research* 17(68):1–30, 2016.

[12] D. Blackwell. Equivalent comparisons of experiments. *Annals of Mathematical Statistics* 24(2):265–272, 1953.

[13] R. Radner, J. E. Stiglitz. A nonconcavity in the value of information. In M. Boyer, R. Kihlstrom (eds.), *Bayesian Models in Economic Theory*, North-Holland, 33–52, 1984.

[14] H. Chade, E. E. Schlee. Another look at the Radner–Stiglitz nonconcavity in the value of information. *J. Economic Theory* 107(2):421–452, 2002.

[15] T. Soleymani, J. S. Baras, K. H. Johansson. Relation between value and age of information in feedback control. arXiv:2403.11926, 2024.

[16] A. Maatouk, S. Kriouile, M. Assaad, A. Ephremides. The age of incorrect information: a new performance metric for status updates. *IEEE/ACM Trans. Networking* 28(5):2215–2228, 2020.

[17] O. Ayan, M. Vilgelm, M. Klügel, S. Hirche, W. Kellerer. Age-of-information vs. value-of-information scheduling for cellular networked control systems. In *Proc. ACM/IEEE ICCPS*, 2019.

[18] T. Soleymani, J. S. Baras, S. Hirche. Value of information in feedback control: quantification. *IEEE Trans. Automatic Control* 67(7):3730–3737, 2022.

[19] W. Bialek, I. Nemenman, N. Tishby. Predictability, complexity, and learning. *Neural Computation* 13:2409–2463, 2001.